

Review: Textual interpretation of transient image classifications from LLMs

Stoppa, F., Bulmus, T., Bloemen, S., Smartt, S. J., Groot, P. J., Vreeswijk, P. & Smith, K. W. (2025), “Textual interpretation of transient image classifications from large language models”, *Nature Astronomy* 9, 1869-1878, doi:10.1038/s41550-025-02670-z.

Summary

Gemini 1.5-pro-002, prompted with 15 annotated new/reference/difference triplets per survey plus a written rubric, is applied to real-versus-bogus classification on Pan-STARRS (2,000 candidates), MeerLICHT (~3,200) and ATLAS (2,000). Accuracy averages 93% (Table 1), each candidate carrying a free-text morphology description and an “interest score” (high/low/no) for triage. The text itself is evaluated twice: 12 astronomers scored explanation coherence 0-5 on 200 MeerLICHT transients, and Gemini scored its own on the same scale. Self-assigned coherence separates Gemini’s correct calls from its incorrect ones (Fig. 4), and using it to pick a few extra exemplars raised MeerLICHT accuracy 93.4% -> 96.7%. A six-month re-analysis quantifies run-to-run and exemplar-set variability, and documents silent drift behind a version-stable endpoint.

Main contributions

1. **A credible few-shot vision-language baseline across three very different cameras.** With 15 exemplars and no training, Gemini reaches 91.9-94.1% accuracy on detectors whose pixel scales span a factor of seven (0.25”, 0.56”, 1.8”).

Survey	Pixel scale	Accuracy	Precision	Recall	Implied FPR
Pan-STARRS	0.25”	94.1%	95.4%	93.1%	~4.8%
MeerLICHT	0.56”	93.4%	87.7%	98.7%	~10.7%
ATLAS	1.8”	91.9%	88.5%	94.5%	~10.3%

(Final column is mine: $FPR = R(1-P)/P \times p/(1-p)$, with p solved from each row’s accuracy - 0.52, 0.44, 0.46, matching the ~50% real fraction stated for ATLAS and Pan-STARRS.)

2. **A measured data-efficiency curve for prompting.** On MeerLICHT, relative to 15 exemplars, 12 costs ~0.5 percentage points, 6 costs ~1.4 pp and 3 costs ~6.9 pp; the ablation is not repeated for the other surveys. The curve is flat above ~12 and collapses below ~6 - a number a deployment can budget against.
3. **Explanation quality measured by domain experts, not by anecdote.** The 12-astronomer panel returned a mean coherence above 4 on 200 MeerLICHT transients, with complete agreement on roughly 120 of them and a median inter-rater standard deviation of 0.6 on the rest.
4. **Self-coherence as an actionable error signal.** Misclassified candidates receive systematically lower self-assigned coherence scores, on both the 200-object panel subset and the full MeerLICHT set (Fig. 4). Feeding low-coherence cases back lifted accuracy 93.4% -> 96.7%: the model’s own uncertainty channel nominated the extra *in-context* exemplars, with a human curating the shortlist rather than combing 3,200 candidates.
5. **An honest repeatability and drift audit.** Five non-overlapping 15-triplet sets matched to the original exemplar class composition x 5 runs (25 inference batches) give within-set sigma $3.0e-4$ (accuracy) and between-set sigma $2.3e-3$. Over six months the same `gemini-1.5-pro` endpoint moved accuracy 0.934 -> 0.962, precision 0.877 -> 0.929, recall 0.987 -> 0.992 - 12x the exemplar-choice scatter and ~90x the sampling scatter. The authors say plainly that periodic revalidation must be planned for.

Critical assessment

The most serious weakness is label provenance. For ATLAS and Pan-STARRS only the $\sim 50\%$ real half was human-classified; the bogus half was “taken from the garbage lists generated by the respective telescope pipelines”, which the paper does not describe. The negative class is therefore *defined* by the incumbent pipeline: part of the reported accuracy measures agreement with that pipeline rather than truth, and any real transient it discarded is scored against Gemini. MeerLICHT labels were assigned by expert astronomers, which is cleaner, but carry no inter-grader statistics.

Second, there is no head-to-head CNN. The abstract claims Gemini “can approach the performance level of a convolutional neural network,” but compares only to literature values (“CNNs commonly exceed 98%”) on other splits. For MeerLICHT the labels are human and the published CNN trained on that same sample (MeerCRAB, ref. 11) is runnable, so a same-split confusion matrix was within reach and is not reported. The gap is 4.6 pp - roughly 150 extra errors on 3,200 candidates, $\sim 460,000$ per night at Rubin scale.

Third, the operating point is unrealistic. The test sets are near-balanced, whereas Rubin, the motivating case, produces ~ 10 million alerts per night at a real:bogus ratio far below unity. At a 1:1,000 prior, MeerLICHT’s 98.7% recall and my derived 10.7% FPR give precision $0.987/(0.987+107) = 0.9\%$; even at 1:100 it is 8.4%. Follow-up is allocated on precision, and the paper gives one threshold, no ROC and no abstention option.

Fourth, the 93.4 $\rightarrow$ 96.7 augmentation gain is not shown to be held out: the extra exemplars came from low-coherence MeerLICHT cases and the model was re-evaluated on MeerLICHT, with no statement that they were excluded. Treat 3.3 pp as an upper bound.

Fifth, the coherence study measures description fidelity, not decision quality. Raters judged whether the text matched the pixels but never classified the candidates independently, so no human accuracy ceiling exists; and no chance-corrected agreement statistic (kappa, QWK) is reported - with the distribution piled on 4 and 5, the 120/200 unanimity rate is close to what chance delivers. Nor is faithfulness tested: nothing shows the stated reason produced the label. And the claimed benefit of querying classifications “as if navigating an annotated catalogue” (abstract) is only sketched in the Discussion, never demonstrated or timed.

To the authors’ credit, several limits are stated squarely: seconds of latency per query against CNN milliseconds, thousands of dollars per night at Rubin scale, prompt-format sensitivity, and the reproducibility hazard of commercial endpoints. One structural caveat: the rubric hard-codes the decision procedure (“The source at the centre can never be negative in the New or Reference image, only in the Difference”), making this a vision-grounded expert rule set more than independent reasoning.

Relation to the agentic lensing programme

- (i) *Self-coherence versus our dead abstention channel.* Gemini’s self-rated coherence separates correct from incorrect calls across the full $\sim 3,200$ -candidate MeerLICHT set (Fig. 4, right) and is worth 3.3 pp when fed back. Our equivalent channel is dead: uncertain fired on 5 of 1,050 three-persona JWST judgements, a diagnosed defect, and LensJudge’s imaging modes sit at $p_{\text{lens}} \sim 0.02\text{--}0.03$ even on consensus-A objects. They never evaluate coherence as a ranker; we do, so the test is cheap - AUC of self-coherence against correctness, benchmarked on the advocate’s 0.764.
- (ii) *Their expert panel is what our labels lack, applied to the wrong quantity.* 120/200 unanimous and median rater sigma 0.6 look nothing like DESI-vs-SuGOHI QWK 0.29 [0.12-0.44], DESI-vs-Euclid 0.17, or Rojas+2023’s 73% repeat-identical. Coherence-of-description is objectively

checkable; a lens grade is not.

- (iii) *Targeted augmentation versus our null results on effort.* Read their 3.3 pp gain against our two nulls: compute buys nothing (lean Sonnet AUC 0.51 versus panel 0.42, multi-agent 0.38, at 2-4x the cost), while rubric reframing moves only the operating point (escalation 12% -> 68%, gold-tier retention 14% -> 33%, all AUC 0.44-0.51).
- (iv) *Their drift audit indicts our frozen benchmarks.* Six months moved the same endpoint 0.934 -> 0.962, roughly 12x their exemplar-choice scatter ($2.3e-3$) and $\sim 90x$ their run-to-run scatter ($3.0e-4$). Every comparative number we hold is a single snapshot on a commercial endpoint: advocate AUC 0.764, recall@5%FPR 33.3% against 16.7% and 9.5%, Opus 4.8's 12% escalation against Sonnet's 42%. At $\sim \$244$ programme spend a scheduled holdout re-run is a rounding error; without one we cannot separate a model-tier result from a vendor update.

Do this: run their exemplar-count ablation on the evidence-first advocate with a strict design/holdout split. LensMark's single validated Opus x xhigh run produced 12 annotation items - enough to prove the few-shot export path, including one *negative* annotation ("star, NOT a counter-image") - so the first job is curating a bank of at least 15 plus a disjoint design set. Power first: on the ~ 86 -positive / 400-negative holdout, 33.3% recall is ~ 29 objects, so a 3.3 pp effect is ~ 3 objects and invisible - run it only for a tier-sized jump or an enlarged positive set. Near-free second adoption: attach a 0-5 coherence rating to the Xiaosheng-only golden campaign (250 items + 40 repeats), an interpretability metric independent of consensus labels.

Where it will not transfer: the resolution result inverts in our regime. Gemini's accuracy barely moves with pixel scale (91.9% at 1.8" versus 94.1% at 0.25") - suggestive rather than a controlled test, since artefact populations and label provenance differ too - because a cosmic-ray streak or dipole residual is a local, instrument-level pattern. Lens detection is the opposite: the same objects graded at DESI 1.3" versus Euclid 0.1" moved mean p_{lens} 0.14 -> 0.75, and 0.05 -> 0.90 on the grade-C subset, because the "C" was an unresolved lens. Label economics do not transfer either: our features score AUC ~ 0.80 on confirmed labels and ~ 0.51 on the hard consensus pool - same features, different labels. Separately, a balanced, pipeline-defined negative class has no analogue at a $1\text{-in-}10^4$ prior, where FPR control dominates (our D-rule veto, FPR@A-or-B 2.0%). And this is prompted classification, not agentic work: no tools, no code execution, no archive queries, nothing like the JWST run's 5,308 cutouts and 1,281 subagents.

Bottom line

This is a well-executed, modest demonstration that a general multimodal model plus 15 exemplars gets within a few points of purpose-built CNNs on real/bogus while emitting text an expert panel rates above 4/5 for coherence, and its repeatability section in Methods is honest about commercial-endpoint drift. It is weakened by pipeline-derived negatives, no same-split CNN comparison, a balanced test set that hides the false-positive problem at survey scale, and an augmentation gain not demonstrably held out. For our programme the transferable content is the machinery, not the accuracy: self-assessed coherence as a routing signal, exemplar-count ablation as a cheap lever, scheduled revalidation of frozen benchmarks. Its central claim should not be expected to survive the move from a balanced, artifact-level task to rare-object lens grading.